HAT-P-32 b

R-0003

Practice run · workflow W8-validation · record `2026-07-07_HAT-P-32b_partial` · created 2026-07-08T01:22:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458107.7056 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1585 +/- 0.0041
Transit depth (Rp/Rs)^2	2.51 +/- 0.13 [%]
Orbital Inclination (inc)	84.01 +/- 1.65
Airmass coefficient 1 (a1)	1.245 +/- 0.0063
Airmass coefficient 2 (a2)	-0.206 +/- 0.022
Scatter in the residuals of the lightcurve fit is	0.51 %
Best Comparison Star	None
Optimal Aperture	3.95
Optimal Annulus	9.19
Transit Duration (day)	0.0988 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1585 ± 0.0041 vs 0.1489 (deviation 2.34σ)
Duration fit vs published	0.0988 d vs 0.1304 d (deviation 7.53σ)
Mid-time O–C	-7.8 min
Depth SNR	38.7
Residual scatter	0.51 %

Light curve

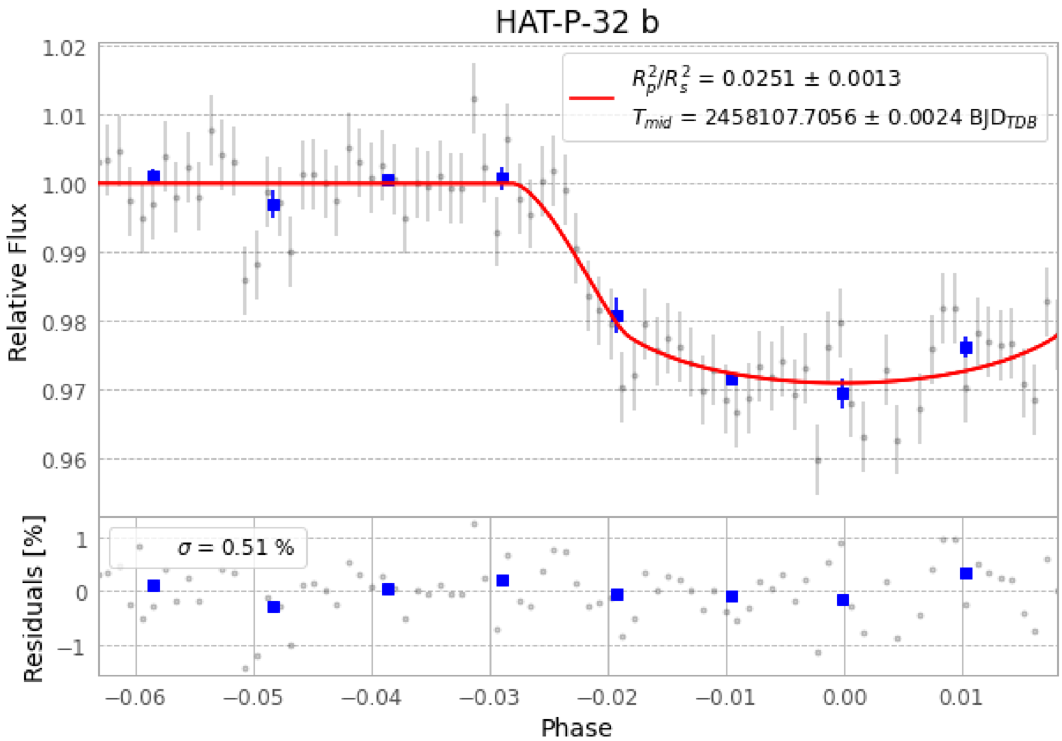

Figure 1 — FinalLightCurve_HAT-P-32 b_17-December-2017.png (SHA-256 4dbeff2067738eeb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 286], 56 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 86ea340b38afe13d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

85 FITS frames (56 MB), `HATP-32171220013343.FITS` → `HATP-32171220054512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: exotic-partial.log (f6eb30421291…), FinalLightCurve_HAT-P-32 b_17-December-2017.png (4dbeff206773…), FinalLightCurve_HAT-P-32 b_17-December-2017.csv (e4752ee94edb…)